



3D vision-based intersecting pipe welding path planning

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Abstract

With the rapid advancement of modern industry, robotic automated welding technology has gradually been applied to the joining of intersecting pipe components. Owing to the stringent quality requirements for such welds and the presence of path deviation errors caused by pipe processing and assembly, conventional off-line programming and “teaching-playback” methods fail to meet the required welding precision. To address these challenges, a 3D vision-based method that integrates 2D images with 3D point clouds has been proposed for planning welding paths of intersecting pipes. First, the YOLO11 and an improved DeepLab V3+ neural network are employed to extract features of the intersecting weld seam. Pixel coordinate mapping is then utilized to obtain the weld seam point cloud. Combined with PCA, the center point of the weld seam is extracted, enabling point cloud reconstruction and fitted interpolation along the entire welding path. Finally, the welding torch posture is planned by integrating the MSAC-WTLSD algorithm with local point cloud information. Experimental results demonstrate that the proposed method enables automatic path planning and torch posture planning for intersecting welds, achieving a welding accuracy within 1 mm.

Highlights

- This study employs 3D vision technology to plan welding paths for intersecting pipes through two-dimensional image processing and three-dimensional point cloud processing. The method described herein enhances robotic welding intelligence by reducing manual intervention while ensuring both efficiency and quality in path planning.
- During two-dimensional image processing, YOLO11 is employed for coarse localization of intersecting welds to enhance the efficiency of subsequent weld extraction. For the weld region of interest (ROI), an improved DeepLab V3+ is utilized to achieve precise weld identification. This approach reduces computational load and increases processing speed while maintaining weld extraction accuracy.
- For welding torch posture planning, a dihedral angle framework is established. By utilizing local point cloud data from the weld seam and integrating the MSAC-WTLSD plane fitting algorithm, the accuracy of plane fitting for sparse point clouds is enhanced. This ultimately enables the determination of the welding torch posture, thereby supporting welding quality control.

Keywords 3D vision · Robotic welding · Weld seam extraction · Welding path planning

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1 Introduction

Intersecting pipe joints are widely employed in large-scale structural components, including bridges, trusses, offshore platforms, and storage tanks. The quality of their welds directly affects the structural reliability of these assemblies. Welding quality is influenced not only by process parameters such as welding speed, current, and voltage but also by the accuracy of trajectory planning for intersecting pipe welds. Unlike straight or curved welds on planar surfaces, the curvature and dihedral angles along intersecting weld curves vary continuously, rendering trajectory planning more complex. Furthermore, in practical production, pipe processing errors and assembly deviations may result in discrepancies between theoretical and physical models [1], which in turn cause deviations between the actual welding path and the robot's predefined trajectory. At present, both the "teaching-playback" [2] and off-line programming [3] approaches are insufficient to satisfy the precision requirements of practical welding applications.

To address these challenges, numerous studies have focused on curved weld seam recognition and trajectory planning using line lasers and two-dimensional image processing. For example, Zou et al. [4] proposed a three-line laser scanning method to detect curved weld trajectories, significantly improving welding accuracy. Jia et al. [5] realized real-time tracking of intersecting welds by integrating laser vision sensors with Kalman filtering and spatio-temporal context algorithms. Yan et al. [6] performed laser prescanning of the welding positions and processed the scanned images algorithmically in combination with locator matching to achieve intersecting curved welding. Zhou et al. [7] developed a weld perception approach based on line structured light scanning, designing corresponding extraction algorithms tailored to weld characteristics. In parallel, deep learning has been widely applied to weld seam identification due to its high accuracy and robustness. For instance, Wu et al. [8] introduced a multi-task detection model based on a CNN architecture, successfully predicting smooth edges and weld seams in large welded components. Wang et al. [9] designed a lightweight network to replace traditional image processing for laser stripe segmentation, enabling precise extraction of weld seam locations. Zhang et al. [10] employed the K-Net model to perform semantic segmentation of welds on irregular workpieces using two-dimensional image features. During welding, the position and orientation of the welding torch also critically affect weld quality. Xu et al. [11] smoothed and interpolated weld feature points to estimate torch pose, while Liu et al. [12] combined theoretical and fitted intersection line models to calculate the welding torch azimuth. Wang et al. [13] further decomposed the torch posture into three independently computed

components—deflection angle, elevation angle, and rotation angle—for comprehensive posture planning. Although these methods can accurately extract weld seams, their practical application in complex environments remains limited [14–16]. Furthermore, they are unable to perform global processing for spatial intersecting welds, and structured-light scanning of weld seams is time-consuming. Consequently, there is a pressing need for a method that combines strong adaptability with an optimal balance between accuracy and efficiency.

In recent years, 3D structured-light vision sensing technology [17, 18] has become increasingly prevalent, and welding path extraction methods based on 3D structured-light vision have received considerable attention. Compared with line structured-light approaches, 3D vision methods obviate the need for manual teaching of the scanning path, enabling more efficient three-dimensional reconstruction and processing of weld seam geometry. For instance, Xiao et al. [19] developed a robotic welding system based on LFoV 3D vision, achieving robust and precise model alignment. Liu et al. [20] proposed a global 3D reconstruction framework combined with local scanning correction using 3D vision perception. Li et al. [21] adopted 3D vision to extract workpiece point clouds, and achieved weld seam positioning and planning via FPFH-RANSAC-ICP algorithm and point cloud segmentation. Geng et al. [22] introduced a 3D vision-based welding path extraction method capable of handling complex intersecting curve welds. Yang et al. [23] achieved extraction of intersecting weld seam point clouds using a lightweight clustering algorithm integrated with an iterative optimization procedure. Zhang et al. [24] developed a 3D vision-driven path planning approach for multi-blade wheel structures, while Liu et al. [25] performed accurate and rapid registration of multi-view point clouds to extract weld seam feature points efficiently, ultimately generating smooth and precise welding trajectories. Despite the advantages of 3D vision in providing wide-field, three-dimensional information for welding path planning, its reliance on global processing can lead to reduced computational efficiency.

In summary, the image processing allows for precise localization of weld seam features, while deep learning improves robustness against disturbances such as reflections and arc light. However, this approach typically requires line-by-line scanning to cover the entire weld seam region, resulting in low efficiency and prolonged processing times. In addition, the scanning path often depends on manual teaching, which limits the level of automation. In contrast, three-dimensional structured-light vision can acquire large-scale point clouds in a single shot, enabling rapid determination of the global spatial orientation of the weld seam. Nevertheless, this method is sensitive to point cloud density and noise, and its feature localization accuracy is generally lower than that

achieved by two-dimensional sub-pixel techniques. Furthermore, processing large volumes of point clouds imposes substantial computational demands. Regarding welding torch posture planning, existing studies predominantly focus on theoretical models, with comparatively less consideration of actual weld seam information.

To address the aforementioned challenges, this paper proposes a novel approach for the extraction and planning of intersecting pipe welds by leveraging the synergy of two-dimensional and three-dimensional information. The primary innovation lies in integrating the rapid and precise feature extraction capability of 2D vision with the spatial processing advantages of 3D point clouds. Specifically, a neural network is employed to extract weld seam features and perform accurate 3D mapping, thereby enhancing the efficiency of point cloud processing. In addition, an MSAC-WTLSF fitting algorithm is introduced to compute the dihedral angles directly from the actual weld seam point cloud, enabling precise welding torch posture planning. This method mitigates path deviations and improves the automation of robotic welding. The total flow chart is illustrated in Fig. 1.

2 Experimental system

As shown in Fig. 2, an experimental platform was established for sensing welding information and planning welding paths. The platform comprises an ABB IRB1410 industrial

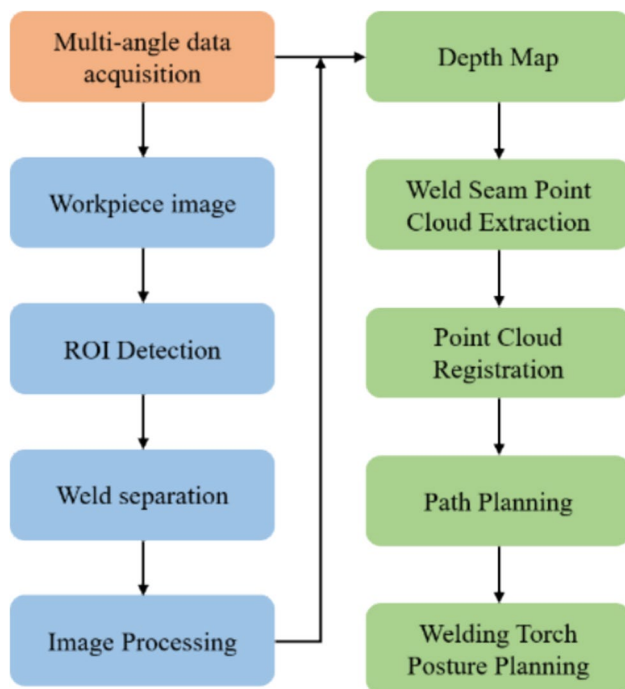


Fig. 1 Total flow chart

robot, Fronius TPS 4000 CMT Advanced automatic welding equipment, and a Mech-Mind NANO ULTRA industrial 3D camera. The welding system utilizes an expert control strategy based on unified welding technology, whereby welding current and voltage are automatically adjusted to match the wire feed speed. The 3D camera is mounted on the welding torch in an Eye-in-Hand configuration, ensuring a fixed relative position between the camera and the torch. Shielding gas consists of a binary mixture of 18% carbon dioxide and 82% argon, and a 1.2-mm solid wire is used as the filler material. The workpiece is an orthogonal intersecting pipe assembly, with a main pipe diameter of 200 mm and a branch pipe diameter of 150 mm, fabricated from Q235 low-carbon steel.

3 Intersecting weld extraction

In the automatic welding of intersecting pipes, accurate extraction of the weld region constitutes the foundation for subsequent path planning and welding torch posture control. However, the weld seam occupies only a small portion of the global image and exhibits pronounced local curvature variations, characterized by multi-scale and densely distributed features. In addition, the grayscale and texture contrast between the weld seam and the pipe surface is relatively weak. Consequently, reliance solely on pixel-level image processing is insufficient for effective weld seam extraction, limiting improvements in segmentation accuracy. To address these challenges systematically, this paper proposes a deep learning-based weld extraction framework, referred to as “coarse localization–fine segmentation.” Within this framework, the YOLO11 model rapidly identifies the region of interest containing the weld seam. Compared with point cloud segmentation, 2D image segmentation offers advantages in terms of lower annotation cost, faster computation, and greater robustness. Building upon rapid localization, an improved DeepLab V3+ model performs sub-pixel-level contour delineation of the weld seam within the local region, emphasizing precise boundary segmentation under complex conditions. Through this synergistic strategy, robust and highly accurate extraction of intersecting pipe welds is achieved, even in challenging industrial scenarios.

3.1 Workpiece image acquisition

To acquire a sufficient number of images of welded workpieces in complex industrial environments, the intelligent vision system used in this study was fully calibrated, including camera calibration, hand-eye calibration, and TCP calibration. The resulting camera parameters are summarized in Table 1. To ensure diversity in the weld seam images, multiple factors were varied during image acquisition, including

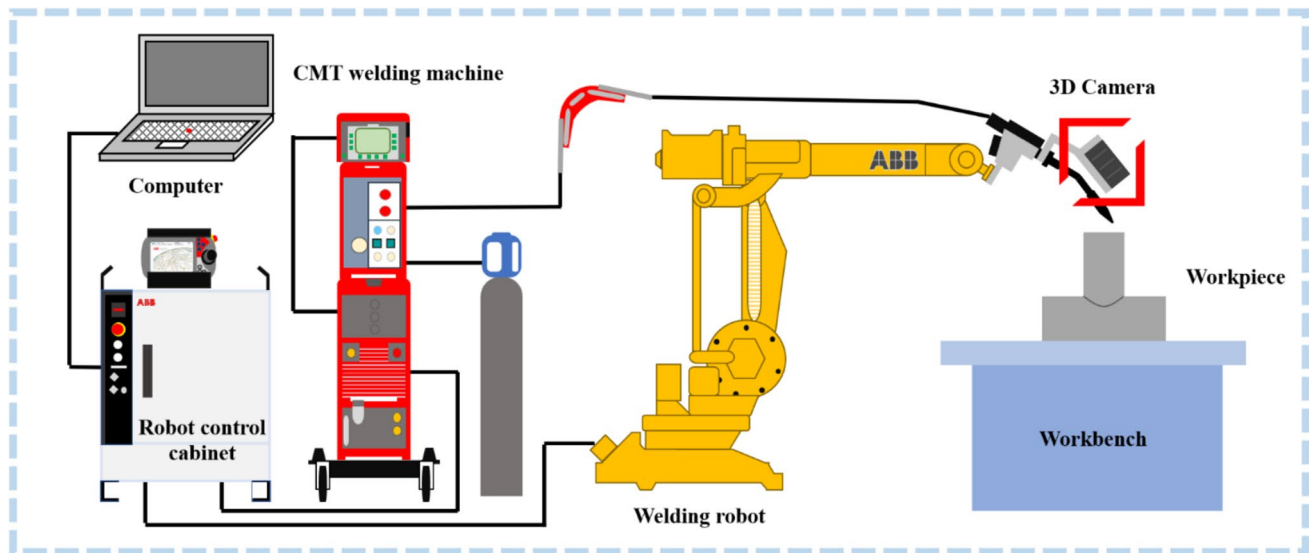


Fig. 2 Schematic diagram of experimental equipment

Table 1 Camera technical specifications

Technical specifications	Specific values
Model	Mech-Eye Nano Ultra
External dimensions	125 × 46 × 76mm
Working distance	250 ~ 500mm
Near field of view	220 × 165 @ 0.25m
Distant field of view	500 × 340 @ 0.5m
Resolution	2400 × 1800
Z-axis repeatability	0.07mm @ 0.5m

workpiece positioning, camera exposure time, shooting angle, workpiece placement, and illumination intensity. The 3D camera captured both grayscale images and corresponding depth maps of the workpiece, enabling subsequent conversion to point clouds.

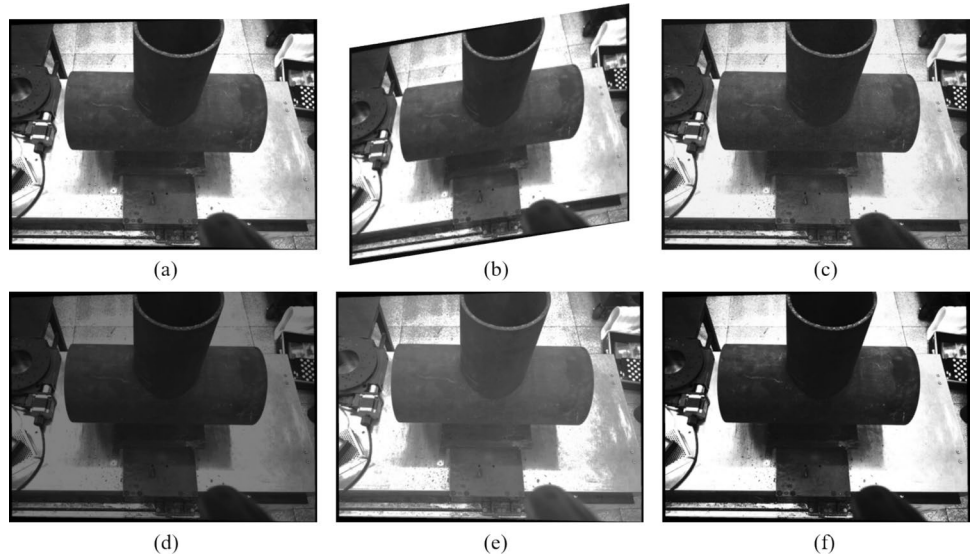
Furthermore, increasing the quantity and diversity of training samples enhances the robustness of the deep learning model. To efficiently expand the dataset, image processing techniques were applied to augment the workpiece images. As illustrated in Fig. 3, the augmentation included (a) the original image; (b) affine transformation, converting the rectangular image into a parallelogram; (c) addition of Gaussian noise; (d) brightness reduction; (e) contrast reduction; and (f) contrast enhancement.

3.2 YOLO11 and the improved DeepLab V3+ neural network

YOLO11 is an image recognition algorithm released in 2024 that builds upon the architecture of previous YOLO versions and incorporates several novel mechanisms. Compared with YOLOv8, enhancements were introduced to both the Backbone and Neck architectures. Specifically, the original C2f module was replaced with the C3k2 module [26], and a new C2PSA module was incorporated to strengthen the model's feature extraction capability. In parallel, the training process was optimized to improve processing speed while maintaining high accuracy, achieving a 22% reduction in parameter count. YOLO11 emphasizes improved detection accuracy without compromising real-time performance, effectively addressing challenges in identifying small, densely packed objects and complex backgrounds [27]. In the context of intersecting pipe welding, YOLO11 provides robust detection of weld regions, supplying reliable Region of Interest (ROI) information for subsequent segmentation.

DeepLab V3+ is a deep learning-based semantic segmentation model [28] that combines an encoder-decoder architecture with Atrous Spatial Pyramid Pooling (ASPP) modules. This design expands the receptive field while preserving spatial resolution, allowing for precise capture of multi-scale contextual information. The decoder

Fig. 3 Augmentation of work-piece images



component further refines object edges, markedly improving boundary segmentation performance. Nevertheless, the model exhibits limitations in handling extremely fine details in complex images, and its performance in segmenting small objects requires further enhancement.

Accurate segmentation of the weld seam is challenging due to the subtle differences between the weld region and the pipe surface, as well as the relatively low image resolution. To address these issues, an improved DeepLab V3+ semantic segmentation algorithm is proposed, targeting difficulties in feature detection and reducing missed detections. In this approach, a lightweight MobileNetV2 [29] replaces the original Xception model, reducing computational complexity while maintaining performance. To further enhance model accuracy by emphasizing target features against complex background information, an ECA-Net module is integrated, enabling the network to balance segmentation accuracy and computational cost without substantially increasing processing time or the number of training parameters. This design reduces information loss and improves the efficiency of local feature extraction. Additionally, the CBAM spatial attention mechanism is incorporated into the decoder to dynamically adjust responses across different channels, thereby enhancing the network's focus on weld seam features and improving feature representation. The combination of ECA-Net and CBAM provides complementary attention mechanisms, achieving a progressive refinement from the “feature dimension” to the “spatial dimension.” The architecture of the improved DeepLab V3+ network is illustrated in Fig. 4.

ECA-Net module ECA-Net (Efficient Channel Attention Network) is a lightweight yet efficient channel attention mechanism [30], as illustrated in Fig. 5. First, global average pooling (GAP) is applied to the input feature map to extract global information from each channel, compressing the feature map to a size of 1×1 . Subsequently, one-dimensional convolutions are introduced to facilitate cross-channel interactions, enabling efficient capture of local channel attention. Finally, the weight for each channel is generated through a Sigmoid function and multiplied element-wise with the original feature map, achieving adaptive recalibration of channel-wise features. The detailed computational procedure is presented in Eqs. (1) and (2). This network enhances the feature extraction capability of deep learning models while introducing only a minimal number of additional parameters, making it well-suited for lightweight computing environments and real-time tasks. By leveraging channel-wise attention, ECA-Net enables the network to automatically learn and strengthen channels that are sensitive to weld features while suppressing interference from irrelevant background information, thereby improving the model's ability to characterize subtle weld features.

$$k = \left\lceil \frac{\log_2 C}{\gamma} + \frac{b}{\gamma} \right\rceil \quad (1)$$

$$\omega = \sigma(C_1 D_k(y)) \quad (2)$$

In the Eq. (1), b and γ are hyperparameters, k represents the dimension, ω denotes the weight value, σ represents the

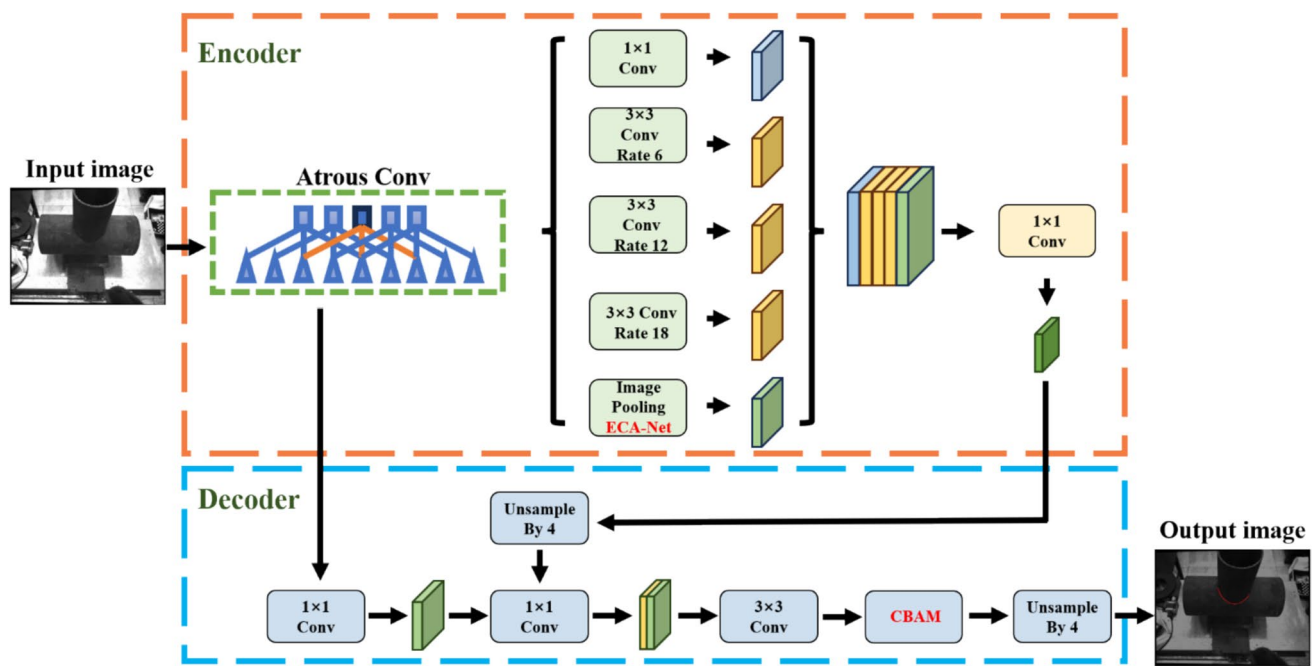


Fig. 4 Improved DeepLab V3+ network architecture

(1) ECA-Net Module

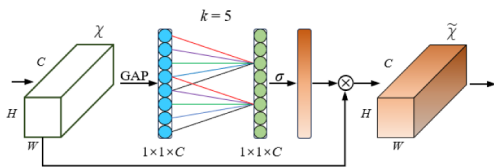


Fig. 5 ECA attention mechanism structure

Sigmoid activation function, $C_l D_k$ denotes a 1D convolution kernel, and y represents the global channel feature of the input.

CBAM module CBAM (Convolutional Block Attention Module) consists of a channel attention module and a spatial

attention module connected in series [31], as illustrated in Fig. 6. This module sequentially attends to both channel and spatial dimensions. In addition to optimizing channel features similarly to ECA-Net, its spatial attention module further enables precise localization of the weld's contour and orientation within the two-dimensional plane. By suppressing spatial responses in background regions, CBAM accurately delineates the specific location and shape of welds in complex spatial environments.

The channel attention module primarily focuses on sharing similar features across different channels and enhances the weights of similar channels, thereby improving the model's understanding of the original image. The channel attention value $M_C(F)$ can be calculated according to formula (3).

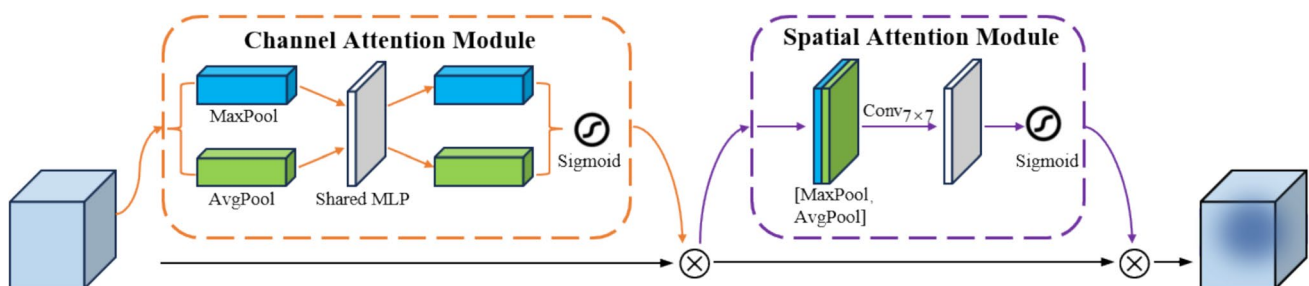


Fig. 6 CBAM attention mechanism architecture

$$Mc(F) = \sigma(MLP(AvgPool(F)) + MPL(MaxPool(F))) = \sigma\left(W_1\left(W_0\left(F_{avg}^c\right)\right) + W_1W_0\left(F_{max}^c\right)\right) \quad (3)$$

F represents the input feature map, σ represents the Sigmoid activation function, MLP refers to the Multi-Layer Perceptron, $AvgPool$ and $MaxPool$ represent global average pooling and max pooling, respectively, W_0 and W_1 are weight parameters, F_{max}^s denotes the average pooling feature map, and F_{max}^c represents the max pooling feature map.

$$M_s(F) = \sigma\left(f^{7 \times 7}\left([AvgPool(F); MaxPool(F)]\right)\right) = \sigma\left(f^{7 \times 7}\left([F_{avg}^s; F_{max}^s]\right)\right) \quad (4)$$

where $f^{7 \times 7}$ denotes a convolution operation with a kernel size of 7×7 , F_{avg}^s represents a 2D feature map from average pooling, and F_{max}^s represents a 2D feature map from max pooling.

The CBAM is applied twice within the network. It processes both the shallow features progressively extracted by the backbone during the encoding stage and the deep features obtained after depthwise separable convolutions and feature fusion, thereby enhancing the model's feature extraction capability and its focus on regions of interest.

Subsequently, the channel attention value $M_c(F)$ is fed into the spatial attention module. This module primarily focuses on the importance of spatial locations within the feature map for the final output, enhancing the weights of nearby positions. The spatial attention value $M_s(F)$ can be computed according to formula (4).

3.2.1 Weld identification results

To construct a dataset for intersecting weld detection, the Labelme tool was employed to label each image. Examples of the original images and their corresponding annotations are shown in Fig. 7. The YOLO11 network was then applied to detect intersecting welds, with the detection results presented in Fig. 8a. The network demonstrated satisfactory performance, meeting the requirements for subsequent

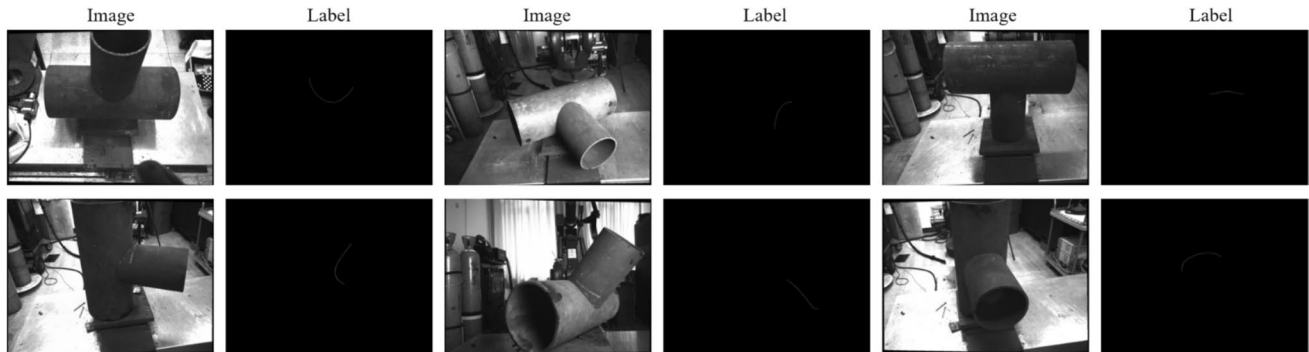


Fig. 7 Part of original image and labeled image

Fig. 8 Weld identification results

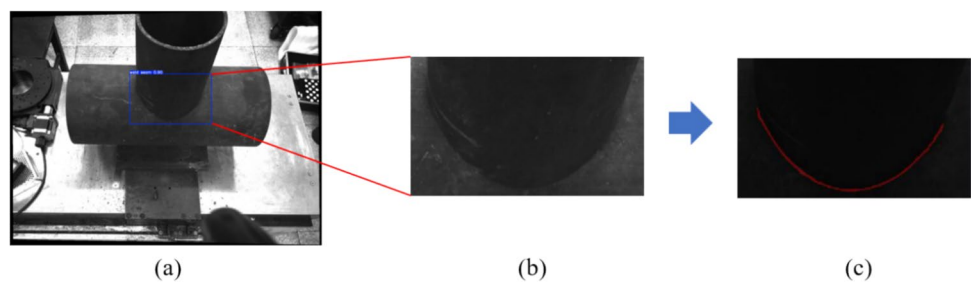


image processing. Pixel coordinates of the four corners of each detection bounding box were extracted, and a rectangle was constructed to define the ROI. The ROI images were subsequently cropped, labeled, and incorporated into the dataset. Finally, the improved DeepLab V3+ network was applied to the cropped images for precise weld segmentation, with the resulting recognition outputs shown in Fig. 8c.

3.3 Image processing of intersecting welds

Since the output images from recognition contain both noise and weld feature information, image processing is a critical step for accurate extraction of weld features. The weld seam predicted by the DeepLab V3+ neural network appears as a red, strip-like region, making manual or empirical determination of the centerline highly challenging and prone to error. To address this, a skeleton extraction algorithm is employed to automatically derive a single-pixel-wide centerline from the predicted weld region. Although this centerline generally lies within the actual weld seam, it may lack sufficient precision and smoothness. Therefore, cubic spline interpolation is applied to enhance accuracy while avoiding overfitting. The specific image processing workflow is illustrated in Fig. 9 and includes the following steps: (a) HSV extraction with Hue intervals of [0,10] and [170,180], Saturation range [70,255], and Value range [50,255]; (b) binarization with a threshold of 30; (c) morphological closing to fill holes using a 10×10 kernel; (d) skeleton thinning via the Zhang–Suen algorithm; and (e) cubic spline interpolation with a sampling density of 2. The resulting curve remains entirely within the predicted weld region and exhibits sufficient smoothness to meet the requirements for subsequent path planning.

4 Intersecting weld point cloud processing

Compared with planar curved welds, spatial intersecting welds exhibit continuously varying curvature and deflection. Although two-dimensional visual techniques offer rapid data processing, they provide only planar contour information. For complex surfaces such as intersecting pipes, 2D images lack depth information and cannot accurately capture the true spatial trajectory and morphology of the weld, posing significant challenges for path planning. The primary advantage of 3D vision lies in its ability to reconstruct the actual three-dimensional trajectory and geometry of the weld with high precision. Paths derived from 3D point clouds can be directly converted into robotic motion trajectories, enabling fully automated operations without the need for manual teaching. Consequently, mapping target pixels from 2D images into corresponding 3D point clouds and processing these point clouds is essential for accurate trajectory planning and automated welding of intersecting pipes.

4.1 Point cloud conversion and preprocessing

A 3D camera projects a structured pattern with specific encoding rules using a projector and captures images of the target surface. Three-dimensional information of the target is obtained through the encoded projection pattern in combination with triangulation methods, allowing the camera to output grayscale images, depth maps, and 3D point clouds. Direct processing and extraction of the full point cloud can be time-consuming and computationally intensive. To mitigate this, the internal parameter matrix of the 3D camera is retrieved via the host computer. By

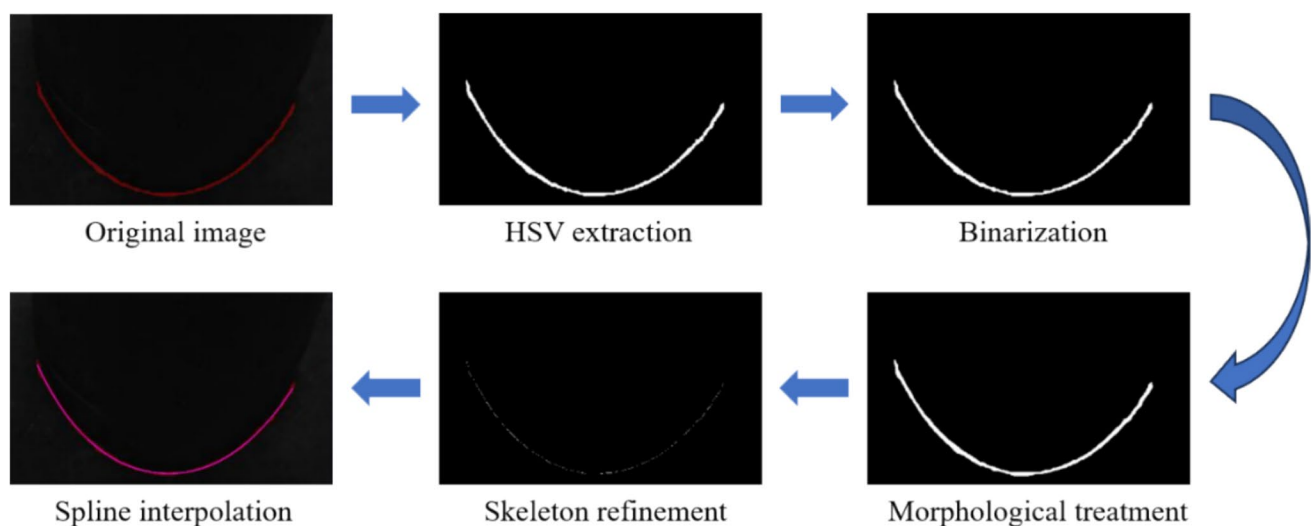


Fig. 9 Image processing flow

integrating the workpiece depth map acquired during data collection with the image processing results described in Sect. 3.3, only the weld feature information is converted into a point cloud. Subsequent processing is performed based on this targeted point cloud, as illustrated in Fig. 10.

During extraction of pixel points from fitting results, noise pixels are inevitably introduced. Similarly, point cloud conversion generates spurious points. To reduce the impact of noise on feature extraction, filtering is applied [32]. Additionally, weld seam regions exhibit significantly higher point cloud density than surrounding areas. Accordingly, Gaussian filtering and statistical filtering are employed to smooth the point cloud while removing sparse regions and outliers.

4.2 Extraction of center points from weld point clouds

After preprocessing, the intersecting weld is represented as a spatial, strip-like point cloud with a finite width. Accurate extraction of the weld centerline from this point cloud is a critical step in weld path planning. Existing centerline extraction methods primarily include curvature analysis, skeleton extraction, and projection techniques. Curvature analysis is suitable for high-quality point clouds but exhibits poor noise tolerance. Skeleton extraction can produce spurious branches and suffers from reduced accuracy due to voxelization. Projection methods are computationally

efficient but are highly dependent on the chosen projection direction and are generally suitable only for coarse extraction. To overcome these limitations, this paper proposes a PCA (Principal Component Analysis)-based slicing method, which adaptively determines the slicing direction, efficiently computes the weld center, and demonstrates high robustness.

PCA is a statistical technique that projects data from a high-dimensional space into a lower-dimensional space while preserving the principal features and structural information. It can effectively identify the main characteristics of the weld, including its orientation and morphology. The detailed procedure of the PCA-based centerline extraction method is described as follows:

- (1) Data centering. First, the mean of each feature is subtracted to center the data at zero. This process can be expressed as:

$$X_{centered} = X - \bar{X} \quad (5)$$

$$\bar{X} = \frac{1}{m} \sum_{i=1}^m X_i \quad (6)$$

where X is the input data matrix of size $m \times n$, with m representing the number of data samples, and n is the number of features per sample, and $\bar{X}$ is the mean vector for each feature.

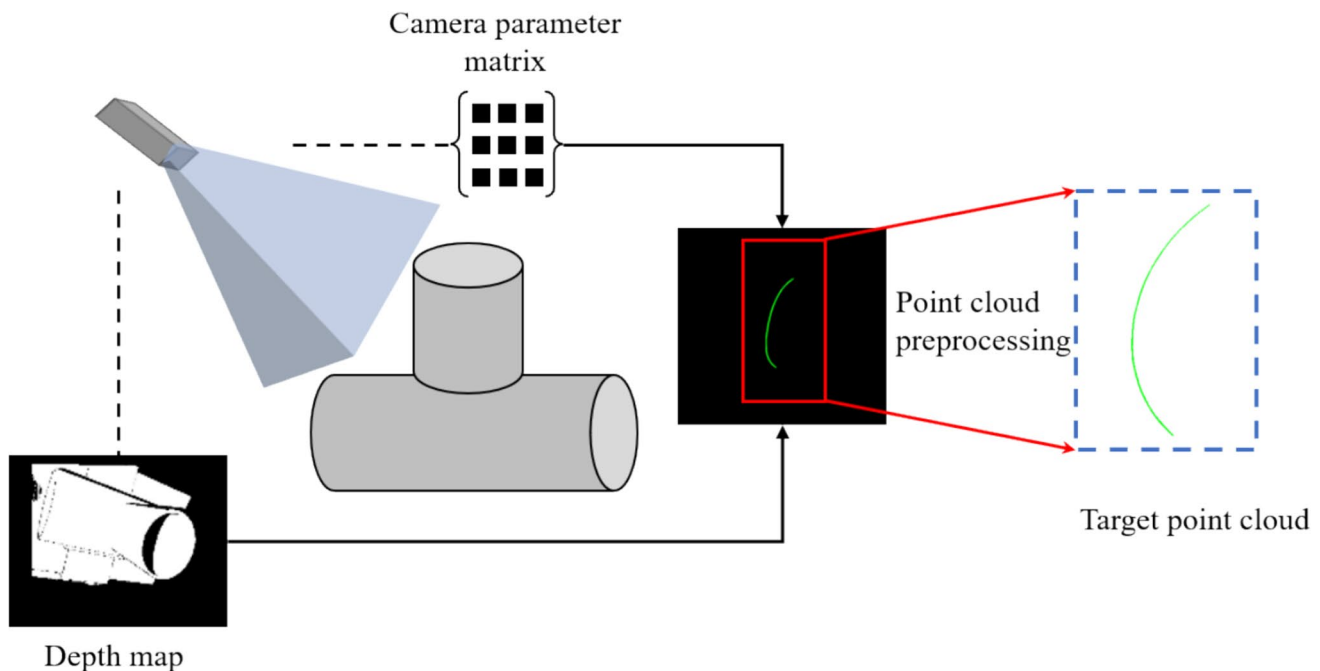


Fig. 10 Schematic diagram of point cloud conversion and processing

- (2) Covariance matrix calculation. The covariance matrix is computed as:

$$C = \frac{1}{m-1} X_{centered}^T X_{centered} \quad (7)$$

C is an $m \times n$ covariance matrix that reflects the correlations among n features.

- (3) Eigenvalue decomposition. Eigenvalue decomposition is performed to solve:

$$C_v = \lambda_v \quad (8)$$

λ is an eigenvalue of the covariance matrix C , and v is the corresponding eigenvector.

- (4) Sort eigenvalues and select principal components. Eigenvectors are sorted in descending order of their corresponding eigenvalues. The eigenvectors associated with the largest k eigenvalues are selected as the principal components. Denote these eigenvectors as the columns of a matrix $V_k = [v_1, v_2, \dots, v_k]$. The original data is then projected onto the principal components to obtain a reduced-dimensional representation:

$$X' = X_{centered} V_k \quad (9)$$

X' is the reduced-dimension data, with dimensions $m \times k$.

- (5) Projection and reconstruction. Projecting the original data onto selected principal components achieves dimensionality reduction. Reconstruct the data using the following formula:

$$X_{reconstructed} = X' V_K^T + \bar{X} \quad (10)$$

As shown in Fig. 11, the overall orientation of the weld seam is determined by computing the principal direction vector and centroid of the weld point cloud using PCA analysis. All points are then projected onto the principal direction to obtain their coordinates along

this axis, thereby defining the start and end points of the weld seam projection. The point cloud is subsequently divided uniformly into multiple slices along the principal direction. Within each slice, all points falling within the interval are counted, and the geometric center of these points is computed to serve as the center point for that slice.

4.3 Weld reconstruction and path planning

To ensure the continuity and integrity of the welding path, locally acquired point clouds must be stitched together. The point cloud stitching process is essentially a coordinate transformation that converts point cloud coordinates from the camera coordinate system to the robot's base coordinate system. First, through hand-eye calibration, the point cloud coordinates in the camera coordinate system are transformed into the robot end-effector coordinate system (i.e., the welding torch coordinate system). Subsequently, using the pose parameters of the robot end-effector, the coordinates are further transformed from the end-effector coordinate system to the robot base coordinate system. Finally, the point cloud coordinates are converted into the base coordinate system, completing the three-dimensional (3D) reconstruction of the intersecting pipes. As expressed in Eq. (11), the weld seam point cloud coordinates from different camera coordinate systems can be transformed into the robot base coordinate system and then stitched together to achieve full 3D reconstruction of both the workpiece and the weld seam. However, overlapping sections may exist in the reconstructed weld seam point cloud. To address this, uniform sampling is performed by setting an appropriate voxel size. In this study, data acquisition and processing are carried out from four distinct viewing angles. The resulting 3D reconstruction outcomes are presented in Figs. 12b and .

$$X_{base} = \left(X_{camera} \cdot {}^{camera}_{top} R_1 + {}^{camera}_{top} T_1 \right) \cdot {}^{top}_{base} R_2 + {}^{top}_{base} T_2 \quad (11)$$

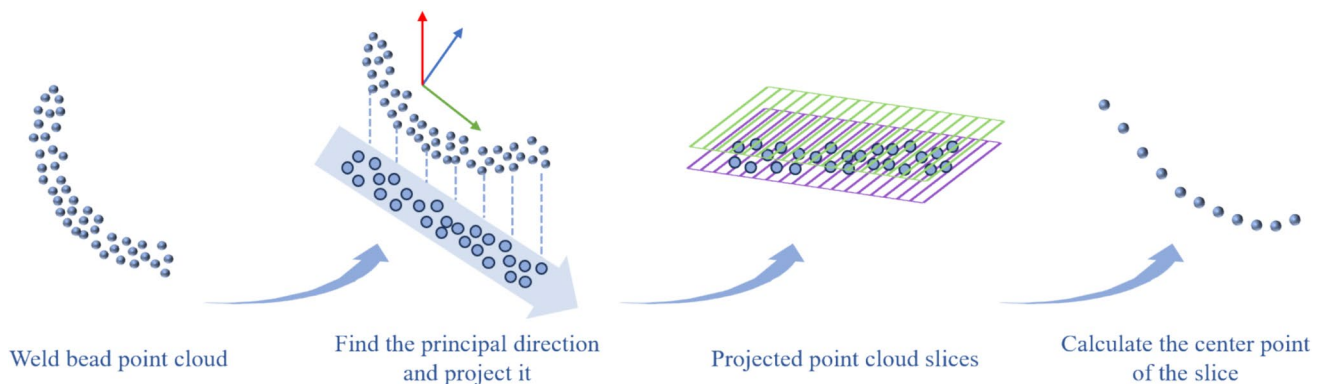


Fig. 11 PCA processing flow chart

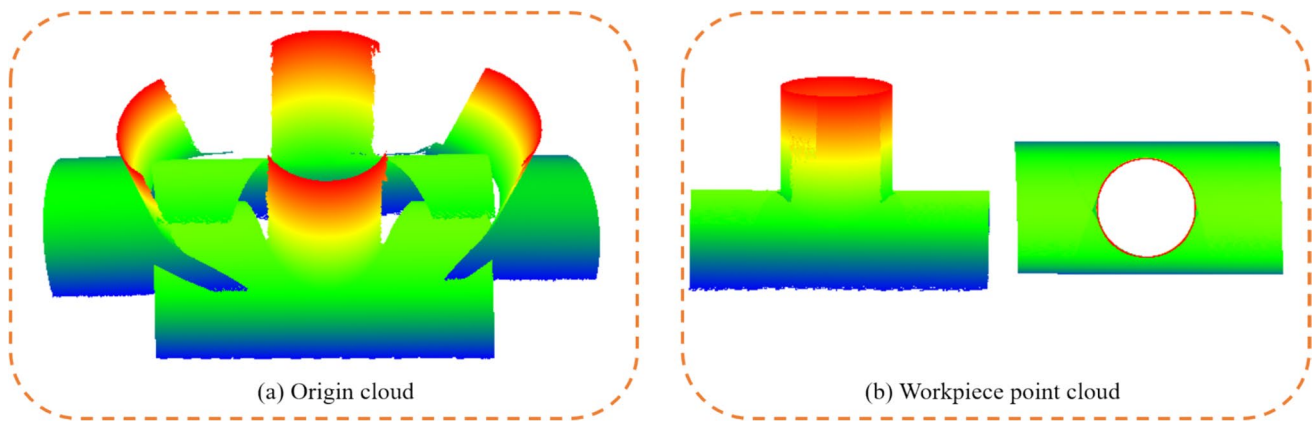


Fig. 12 Point cloud reconstruction results for intersecting pipe components

where X_{base} represents the point cloud coordinate matrix transformed to the base coordinate system, X_{camera} represents the point cloud coordinate matrix in the camera coordinate system, ${}^{camera}_{top}R_2$ and ${}^{camera}_{top}T_2$ represent the rotation and translation matrices for transforming from the camera coordinate system to the robot end-effector respectively, and ${}^{top}_{base}R_2$ and ${}^{top}_{base}T_2$ represent the rotation and translation matrices for transforming from the robot end-effector to the robot base coordinate system, respectively.

To ensure smooth robot motion, a NURBS interpolation method is employed. This approach provides a smooth estimation of the weld seam, more accurately representing its actual geometry. It also reduces abrupt changes and vibrations during robotic movement, thereby enhancing both stability and precision in the welding process. In this study, cubic NURBS fitting and interpolation are utilized to

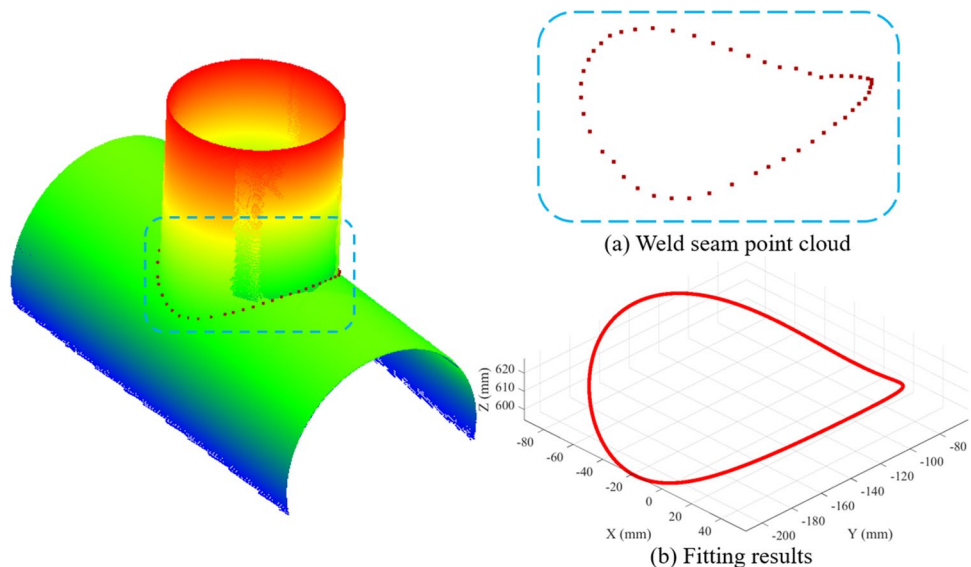
maintain high fitting accuracy while avoiding overfitting. The resulting weld trajectory is illustrated in Fig. 13b.

4.4 Welding torch posture planning

Unlike welding on flat or simple curved surfaces, welding intersecting joints requires careful consideration of welding torch obstacle avoidance. Locally, the structure forms a dihedral framework defined by two curved surfaces and their intersection line, with the dihedral angle varying along the weld position. Constructing a dihedral angle framework for the intersecting weld enables precise planning of the welding torch posture throughout the welding process.

As illustrated in Fig. 14, O_F denotes a point on the intersecting weld. Using this point as the origin, a weld local coordinate

Fig. 13 Weld path fitting interpolation results



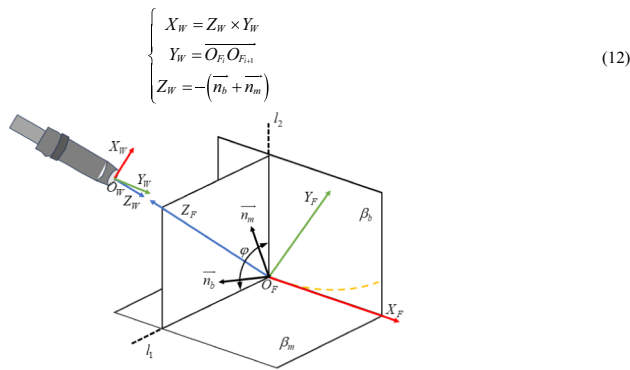


Fig. 14 Schematic diagram of torch posture

system $\{F\}$ is established, β_b and β_m represent the tangent planes of the branch pipe and main pipe at point O_F with unit normal vectors $\vec{n}_b$ and $\vec{n}_m$ respectively. The angle φ between these two normal vectors constitutes the current dihedral angle, optimal welding results are achieved when the direction of the welding torch's central axis aligns with the bisector of this dihedral angle, simultaneously minimizing collision risks between the torch and the pipe.

Assuming that in the initial posture, the Y -axis direction of the welding torch coordinate system $\{W\}$ is aligned with the tangent direction of the current weld point, which corresponds to the X -axis of the weld coordinate system. When interpolation points along the welding trajectory are densely distributed, applying differential concepts allows the tangent direction to be represented as the line segment connecting the current and next welding points, using the weld point as the sphere's center, select points within a radius r for planar fitting to determine the unit normal vectors $\vec{n}_b$ and $\vec{n}_m$. Employing the dihedral angle principle, calculate the Z_F -axis of the welding torch. The X -axis of the torch is then obtained via the cross product of the Z - and Y -axes. Thus, the welding torch coordinate system $\{W\}$ is constructed as follows:

$$\begin{cases} X_W = \frac{Z_W \times Y_W}{|Z_W \times Y_W|} \\ Y_W = O_{Fi} O_{Fi+1} \\ Z_W = -(\vec{n}_b + \vec{n}_m) \end{cases} \quad (12)$$

During point cloud acquisition, factors such as equipment vibration, environmental disturbances, and uneven workpiece surfaces inevitably introduce noise and outliers. To mitigate the impact of erroneous points on plane fitting, this study proposes a weighted fitting approach that combines the MSAC (M-estimator Sample Consensus) algorithm with the WTLSD (Weighted Total Least Squares Based on Distance) method. Specifically, the MSAC algorithm is first applied to perform an initial fit of the point cloud. Subsequently, a distance-based weighted total least squares plane fitting is conducted to refine the result, yielding the final fitted plane.

(1) MSAC algorithm.

The MSAC (M-estimator Sample Consensus) algorithm is a robust fitting method and an improved variant of RANSAC [33]. By modifying the cost function, it reduces sensitivity to threshold values and accelerates convergence speed. Primarily employed for estimating model parameters from datasets containing noise and outliers, the algorithm essentially seeks to minimize the cost function H , defined as follows:

$$H = \sum_i^n \rho(e, T) \quad (13)$$

n is the total amount of data, i represents the sequence number of the data, and $p(e, T)$ is the cost function.

The cost function is expressed as follows:

$$\rho(e, T) = \begin{cases} e, & e \leq T \\ T, & e > T \end{cases} \quad (14)$$

When the distance e between a point and the model is less than the threshold T , the point is classified as an interior point with weight e ; otherwise, it is classified as an exterior point with weight T .

The fundamental principle of the MSAC algorithm for fitting planar models is as follows:

- 1) From the point cloud dataset $\mathcal{Q}$, randomly select the smallest sample set $\mathcal{C}$ without replacement that satisfies the requirements of the fitted plane model.
- 2) Estimate the plane parameters corresponding to the sample set $\mathcal{C}$ using the least squares method, and take these as the initial model parameters.
- 3) Calculate the distance from all points in $\mathcal{Q}$ to the initial plane model. Points whose distances are below a predefined threshold T are considered inliers, then, compute the cost C_I of this model.
- 4) Compare the cost values, retain the smaller one as the cost of the new best model, and record the corresponding inlier points and model parameters.
- 5) resulting in the final set of inlier points. Using these inlier points, fit a plane model via the least squares method to obtain the optimal plane model parameters.

(1) WTLSD algorithm.

The point cloud data plane expression is expressed as follows:

$$z_i = ax_i + by_i + c \quad (15)$$

a, b, c denote the parameters to be determined in the plane equation. An errors-in-variables (EIV) model is established to account for both the measurement errors inherent in the observations and the errors present in the model coefficient matrix:

$$\mathbf{Z} - \mathbf{e} = (\mathbf{A} - \mathbf{E})\xi \quad (16)$$

where $\mathbf{Z}$ represents an $n \times 1$ observation vector, $\mathbf{e}_z$ denotes the random error of the observation vector $\mathbf{Z}$, $\mathbf{A}$ represents an $n \times 3$ coefficient matrix, $\mathbf{E}_A$ represents the random error of the coefficient matrix, ξ is the coefficient to be determined in Eq. (15).

The mean vector and covariance equation for the error vector $\mathbf{e}_z$ and $\text{Vec}(\mathbf{E}_A)$ are expressed as follows:

$$\begin{bmatrix} \mathbf{e}_z \\ \text{Vec}(\mathbf{E}_A) \end{bmatrix} = \begin{bmatrix} \mathbf{e}_z \\ \text{Vec}(\mathbf{E}_A) \end{bmatrix} \sim \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \sigma_0^2 \begin{bmatrix} \mathbf{Q}_z & 0 \\ 0 & \mathbf{Q}_A \end{bmatrix} \right) \quad (17)$$

$\mathbf{Q}_z$ and $\mathbf{Q}_A$ represent the non-singular covariance matrices of error vectors $\mathbf{e}_z$ and $\mathbf{E}_A$, respectively, with pre-test unit variance σ_0^2 . The parameter estimation criterion for WTLS is expressed as follows:

$$\min(\mathbf{e}_z^T \mathbf{Q}_z^{-1} \mathbf{e}_z + \mathbf{e}_A^T \mathbf{Q}_A^{-1} \mathbf{e}_A) \quad (18)$$

In the WTLS algorithm, the weighting principle of the weighting matrix is determined based on the distance between the three-dimensional coordinates of points in the point cloud and the fitted plane. The relevance of a given point to the predicted plane is evaluated by calculating the distance from the point to the plane. The row vector matrix is defined as follows:

$$\mathbf{P}_{XY} = \text{diag}[\mathbf{P}_1] \quad (19)$$

$$\mathbf{P}_Z = \text{vec}[\mathbf{P}_1] \quad (20)$$

The column vector weight matrix of the coefficient matrix is expressed as follows:

$$\mathbf{P}_0 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (21)$$

Initial plane fitting is performed using Total Least Squares (TLS). Subsequently, the inverse of the distance from each point to the initial plane is calculated as the weight for the initial weight matrix. Points farther from the plane receive smaller weights, while those closer receive larger weights. The initial weight calculation method is as follows:

$$\mathbf{P}_{i0} = l_{i0} \quad (22)$$

$$\begin{cases} l_i = \sqrt{\frac{|ax_i + by_i - z_i + c|}{a^2 + b^2 + 1}} \\ l_{i0} = \frac{|l_i - l_{\min} + \phi|}{l_{\max} - l_{\min}} \end{cases} \quad (23)$$

The MSAC-WTLS method first conducts an initial screening of the point cloud using the MSAC algorithm to fit a preliminary plane with reasonable accuracy. Subsequently, the WTLS algorithm iteratively refines this initial plane to mitigate the influence of defective regions on the final result, as illustrated in Fig. 15. The parameter calculation procedure is as follows:

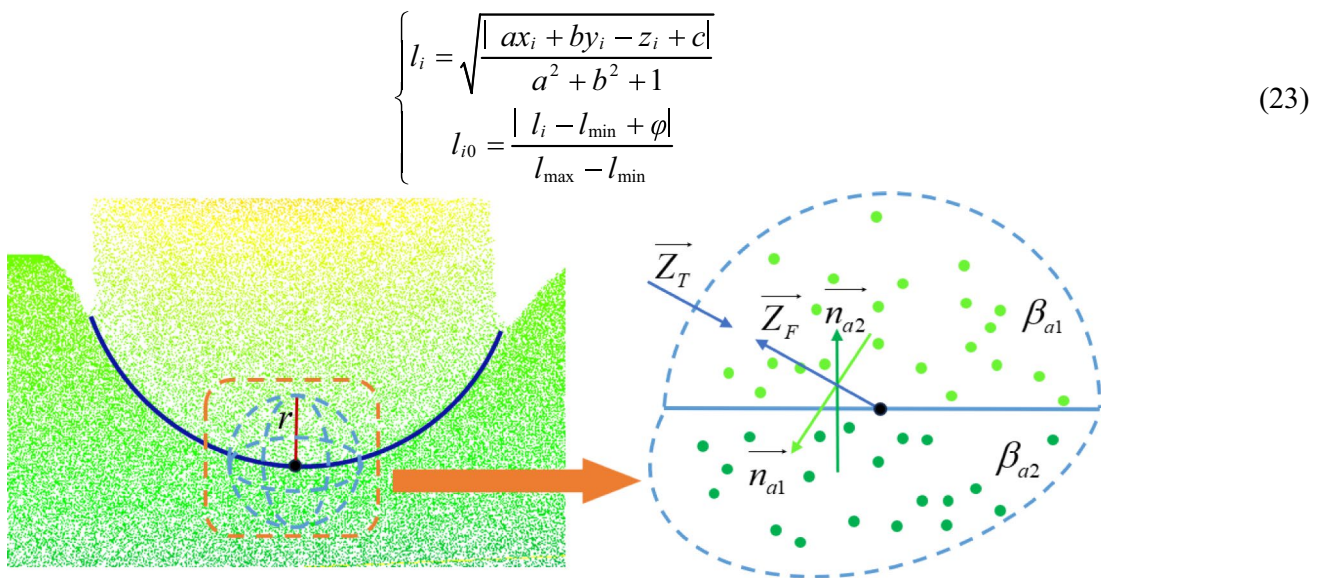


Fig. 15 Schematic diagram of local point cloud fitting

- 1) The plane parameters are estimated using the MSAC algorithm. Subsequently, the column vector weight matrix P_0 of coefficient matrix A, the initial row vector weight matrix $P_{XY(0)}$, and the initial observation value matrix $P_{Z(0)}$ are established.
- 2) Calculate the iterative initial values for the plane parameters $\xi = [a, b, c]^T$:

$$\begin{cases} \hat{v}_{(0)} = 0 \\ \hat{\xi}_{(0)} = (A^T Q_{Z(0)} A)^{-1} A^T Q_{Z(0)} Z \\ \mu_{(0)} = \left(Q_{Z(0)} + \left(\left(\hat{\xi}_{(0)} \right)^T Q_0 \hat{\xi}_{(0)} \right) Q_{XY} \right)^{-1} \\ \hat{\xi}_{(1)} = (A^T \mu_{(0)} A)^{-1} A^T \mu_{(10)} Z \end{cases} \quad (24)$$

- 3) Update the weight matrix P_{XY} , P_Z by calculating the distance between the point cloud data and the iterated plane.
- 4) Calculate $\hat{\lambda}_i$, $\hat{v}_i$, and $\hat{\xi}_{(i+1)}$

$$\begin{cases} \hat{\lambda}_{(i)} = \mu_{(i)} \cdot (Z - A \hat{\xi}_i) \\ \hat{v}_{(i)} = \left(\hat{\lambda}_{(i)} \right)^T Q_{XY} \hat{\lambda}_{(i)} \\ \hat{\xi}_{(i+1)} = (A^T \mu_{(i)} A - \hat{v}_i \cdot Q_0)^{-1} \cdot (A^T \mu_{(i)} A) \end{cases} \quad (25)$$

If $\hat{\xi}_{(i+1)} - \hat{\xi}_{(i)} \cdot \delta_0$, repeat the step; otherwise, end the iteration.

The final welding torch posture planning results are shown in Fig. 16.

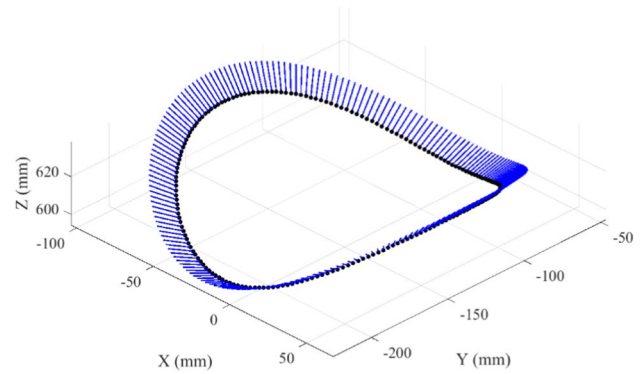


Fig. 16 Welding torch posture planning results

5 Experiments

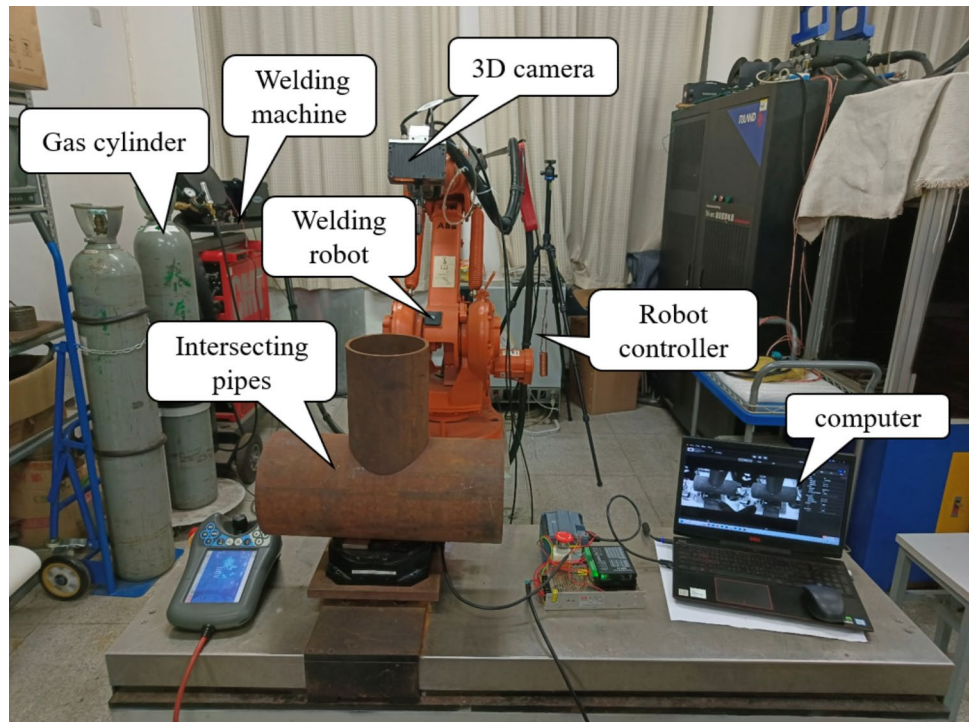
Figure 17 shows the build experiment platform to verify the proposed method, primarily used to validate the model performance and the effectiveness and accuracy of the proposed intersecting weld extraction method.

5.1 Model training and performance testing

5.1.1 Model training

Camera calibration and hand-eye calibration were performed using Mech-Eye View software in conjunction

Fig. 17 Experimental platform



with the calibration board provided with the camera. The TCP calibration of the welding torch was accomplished using the robot's built-in tool functionality. To capture a diverse and representative dataset in complex environments, workpiece position, camera exposure time, shooting angle, workpiece placement pose, and illumination intensity were varied during image acquisition. A total of 750 images containing intersecting pipes were collected, of which 50 were randomly selected for data augmentation. The resulting dataset comprised 1,000 images, which were partitioned into training and test sets at a ratio of 9:1. All test set samples are original images without any repetition or overlap with the training set.

The YOLO11 and improved DeepLab V3+ networks were subsequently trained on this dataset. Training was conducted using the PyTorch framework on a Windows 11 system, with the experimental environment detailed in Table 2.

During training of the YOLO11 model, SGD was employed as the optimizer, with a fixed input image size of

640×640 pixels, a batch size of 16, and a total of 200 training epochs. For the modified DeepLab V3+ network, the Adam optimizer was used, with an input image resolution of 512×512 pixels and an initial learning rate of 0.005. Cosine annealing was applied for learning rate decay. The batch size was set to 4, and training was performed for 200 epochs.

5.1.2 Module performance testing and comparative experiments

Through training the YOLO11 model, it achieved a precision of 95.9%, a recall of 92.2%, and a mean average precision (mAP) of 93.5% on the test set. mAP50% represents the average precision when the intersection over union (IoU) is set to 0.5. Selected detection results are shown in Fig. 18. The results demonstrate that this model can achieve accurate detection of ROIs for intersecting welds.

To evaluate the performance of the improved DeepLab V3+ model, ablation experiments are conducted in this study, along with comparative analyses against existing segmentation models. Several evaluation metrics are adopted, including the number of model parameters (Params), mean intersection over union (MIoU), mean pixel accuracy (MPA), overall accuracy (OA), and frames per second (FPS). The corresponding experimental results are presented in Table 3.

As shown in Table 3, although PSPNet achieves the fastest inference speed, it suffers from a large number of parameters and low segmentation accuracy. U-Net improves detection accuracy; however, its parameter count and inference speed still require further optimization. Replacing the backbone network with MobileNetV2 achieves outstanding model lightweighting while maintaining segmentation precision. The improved ECA module enhances recognition

Table 2 Experimental environment

Name	Content
CPU	Intel(R) Core(TM) i7-9750H CPU @ 2.60GHz
GPU	NVIDIA GeForce GTX 1650(12GB)
CUDA	11.8
PyTorch	2.5.1
Language	Python

Fig. 18 ROI detection

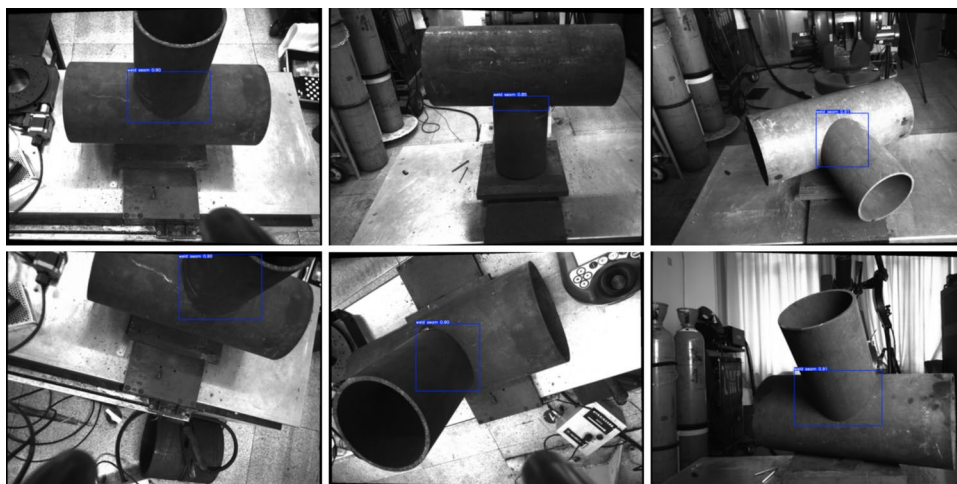


Table 3 Ablation experiment results

Model (backbone network)	Params/MB	Miou/%	MPA/%	OA/%	FPS
PSPNet (ResNet-18)	45.88	80.23	85.12	87.55	62.63
U-net (VGG16)	33.25	90.92	92.09	93.17	54.32
DeepLab V3+ (ResNet-18)	57.43	88.34	91.21	92.94	41.38
DeepLab V3+ (MobileNetV2)	15.34	89.82	91.19	93.51	58.44
DeepLab V3+ +ECA (MobileNetV2)	15.36	90.95	92.48	93.87	57.42
DeepLab V3+ +CBAM (MobileNetV2)	15.48	92.76	94.05	94.21	55.59
Improved DeepLab V3+ (MobileNetV2)	15.51	93.79	95.63	95.96	55.75

capability with almost no increase in model parameters, and the optimization effect of the CBAM module is even more remarkable. The optimized DeepLab V3+ exhibits stronger overall detection performance and improved segmentation ability for complex features. Figure 19 presents the segmentation results obtained by different models. Comparative analysis verifies that the improved model demonstrates superior accuracy and stability in weld seam identification and segmentation.

To evaluate the generalization capability of the trained models, weld extraction experiments were performed on intersecting pipes of two additional sizes, fabricated from PVC. One specimen was an orthogonal intersecting pipe with a main pipe diameter of 110 mm and a branch pipe diameter of 90 mm, while the other was a 45° oblique intersecting pipe with a main pipe diameter of 110 mm and a branch pipe diameter of 75 mm. Partial verification results under varying lighting conditions and different workpiece poses are shown in Fig. 20.

Both models successfully extracted the weld regions with reasonable accuracy across different pipe dimensions, materials, and intersection angles, demonstrating a certain degree of generalization and adaptability to moderate variations. Nevertheless, achieving higher generalization requires retraining with a more diverse dataset. Incorporating multiple oblique intersection angles, as well as multi-pipe intersections of varying sizes, would improve adaptability to geometric variations. Furthermore, expanding the dataset to include extreme lighting conditions and diverse material surfaces is necessary to reduce false detections caused by environmental interference.

5.2 Experimental results and error analysis

In this study, welding path planning was conducted for intersecting pipes at various positions and orientations. The experimental results are presented in Fig. 21, where the black points denote the planned welding points and the blue lines indicate the Z-axis direction of the welding torch.

To assess the accuracy of the proposed method, the welding path obtained via the conventional “teaching-playback” approach was used as a reference to calculate the mean error (ME) and root mean square error (RMSE). A total of 25 welding points were manually taught along the actual welding path and compared with the trajectory curve fitted by the proposed method. As shown in Fig. 22, the red points represent the selected weld points on the trajectory generated by the proposed method, whereas the blue points correspond to the manually taught weld points. Figure 23 presents the errors along the x-, y-, and z-axes between the manually taught points and those obtained using the proposed method. The quantitative results are summarized in Table 4. The average deviation between the proposed method and the manually taught points is less than 1 mm, with an RMSE below 0.9 mm, satisfying the typical welding error tolerance. These results indicate that the welding trajectory generated by the proposed method can effectively replace trajectories obtained through manual teaching.

To verify the consistency of the welding torch attitude planning, the variation θ of the welding torch between adjacent welding points and its rate of change ω along the weld seam are introduced as core evaluation metrics. The calculation formulas are as follows:

$$\theta = \arccos \left(\vec{Z}_f \cdot \vec{Z}_{f1} \right) \times \frac{100}{\pi} \quad (26)$$

$$\omega = \frac{\theta}{\sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2 + (z_{i+1} - z_i)^2}} \quad (27)$$

The calculation results are shown in the Fig. 24, where Fig. 24 (1) represents the angular variation in torch angle and Fig. 24 (2) represents the angular variation rate of the torch angle. The red curve corresponds to Group (a), and the blue curve corresponds to Group (b). As can be seen from the Figure, both datasets exhibit stable periodic fluctuations in the overall trend, without abrupt changes or anomalous deviations. The attitude adjustment process is

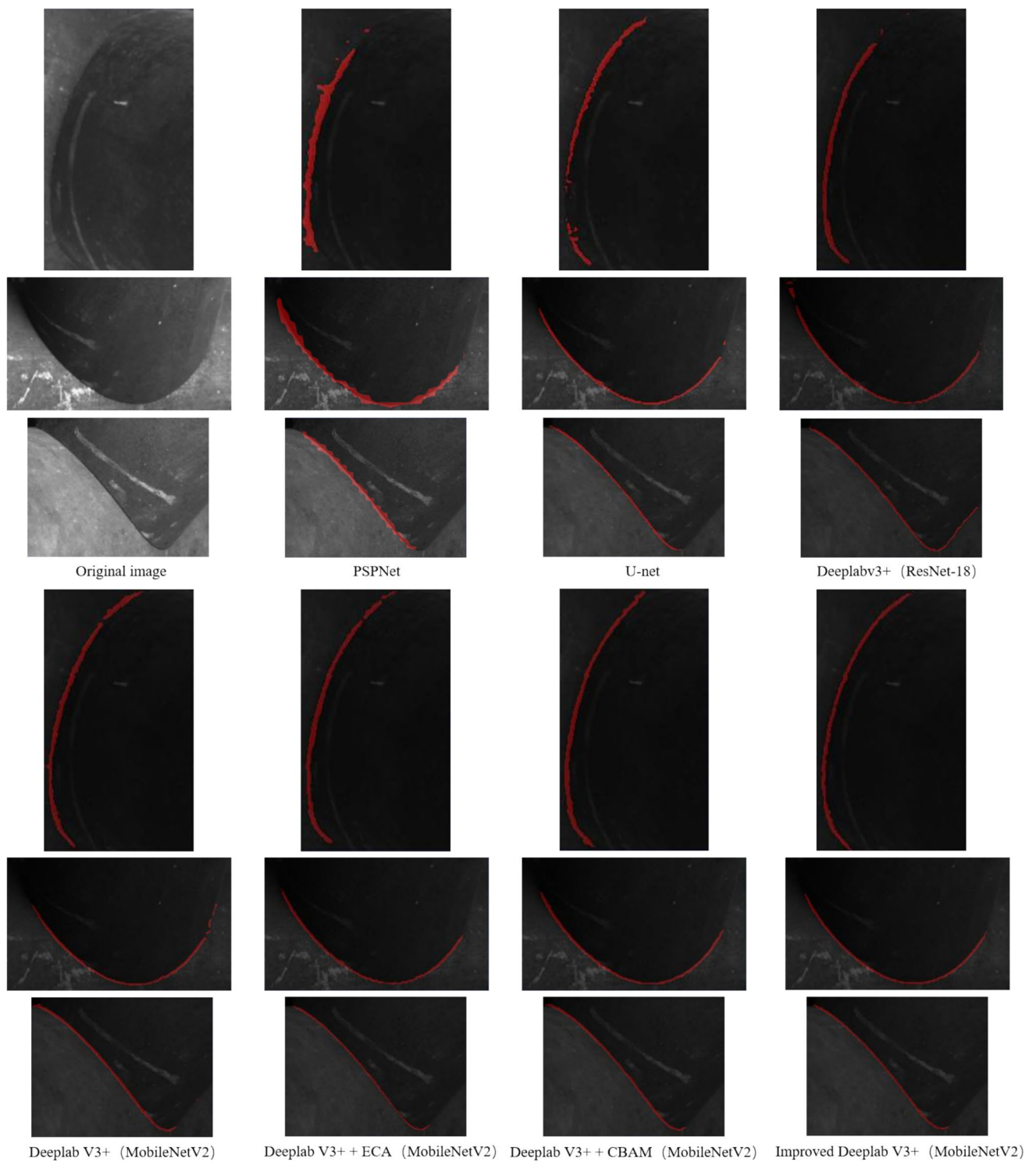
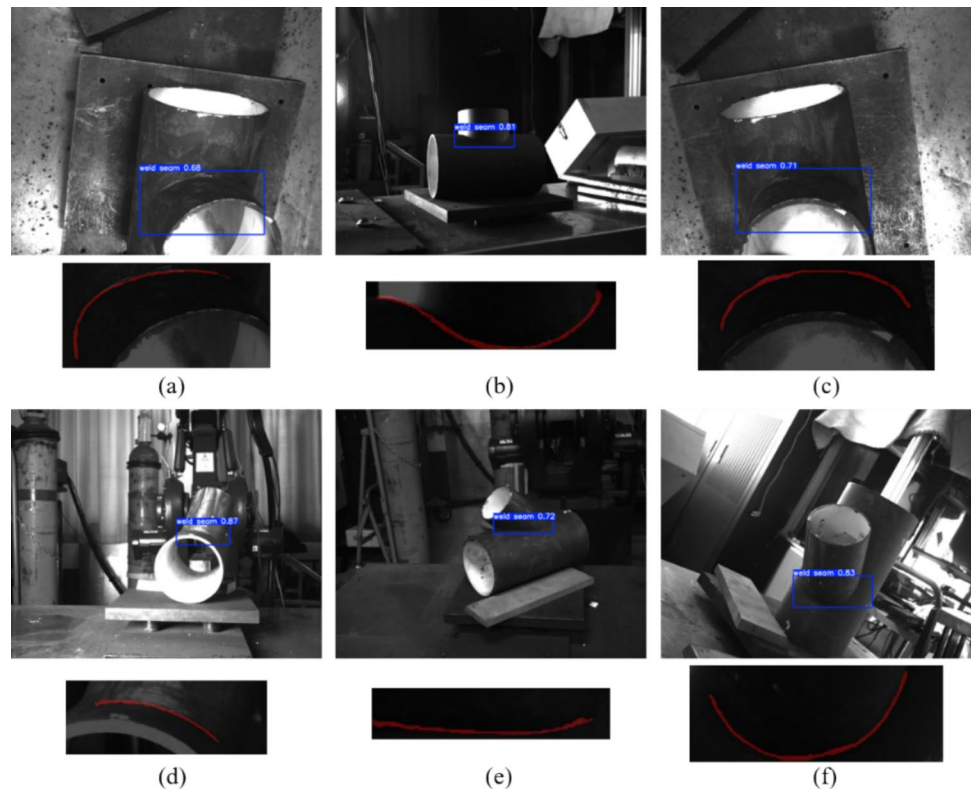


Fig. 19 Intersecting weld identification

Fig. 20 Partial verification results



smooth and continuous, demonstrating excellent consistency in torch attitude.

5.3 Generalized verification

To evaluate the generalizability of the proposed method, the oblique intersecting pipe described in Sect. 5.1.2 was used as a validation object. Data were acquired using a 3D camera, and welding path planning was performed according to the proposed method. The reconstruction results and planned trajectories are presented in Fig. 25. Although the accuracy of weld feature extraction was slightly reduced, successful welding path planning was still achieved. This experiment demonstrates the method's generalizability and further confirms its robustness in planning welding paths for pipes with varying intersection angles.

6 Conclusion

To address deviations between actual and ideal welding paths in intersecting pipe joints caused by machining errors and workpiece assembly inaccuracies, this study proposes a

3D vision-based robotic welding trajectory planning method. The approach enables both extraction and planning of welding paths for intersecting pipes, as well as welding torch pose planning, achieving precision that satisfies practical welding requirements. The proposed method offers several advantages:

- 1) Automated welding path planning is realized, eliminating the need for manual teaching and programming while significantly improving welding efficiency and automation.
- 2) High-quality planar fitting is performed based on the actual weld seam point clouds to optimize the orientation of the welding torch.
- 3) Deviation compensation addresses discrepancies between the actual and ideal welding paths caused by workpiece machining and assembly errors, ensuring the practicality and high precision of the resulting welding trajectory.

The proposed method is applicable not only to orthogonal intersecting curve welds but also demonstrates a degree of broad applicability. Nonetheless, limitations remain: variations in workpiece size and material can affect the

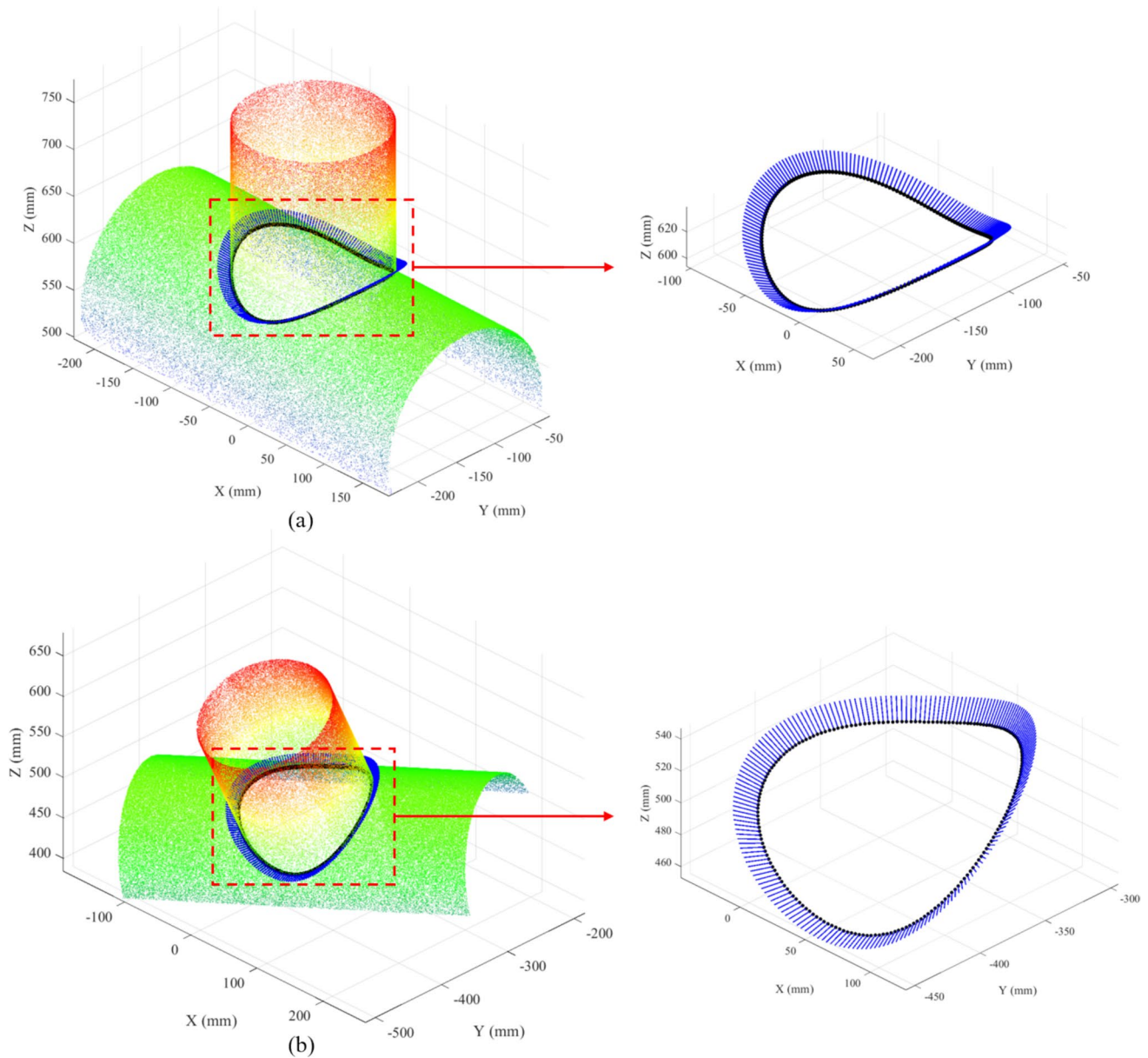


Fig. 21 Experimental results

Table 4 Experimental error results

Case	Error (mm)	x	y	z
1	ME (a)	0.834	0.921	0.986
2	RMSE (a)	0.752	0.683	0.701
3	ME (b)	0.906	0.899	0.989
4	RMSE (b)	0.817	0.765	0.824

accuracy of path planning, and the current experiments were conducted under relatively idealized conditions. Adaptation to more complex welding scenarios requires further investigation. Future work will focus on optimizing the algorithm, expanding the dataset to include a wider variety of intersecting welds and ideally performing actual welding and integrating welding process parameters with weld seam geometric features.

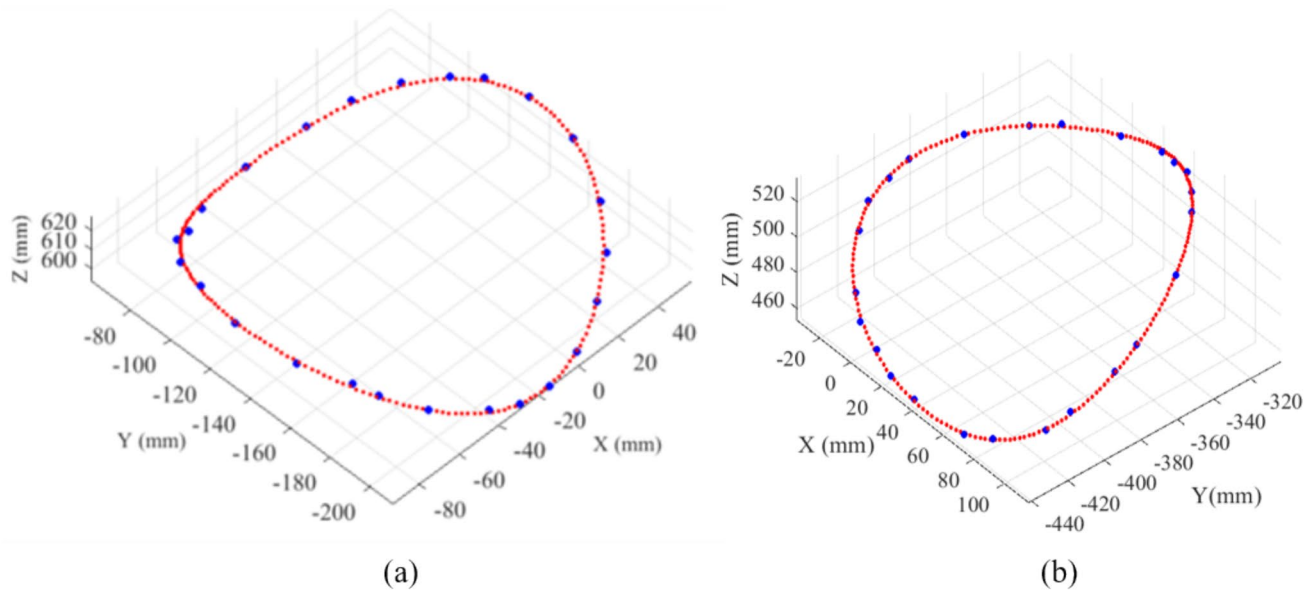


Fig. 22 Manual teaching path

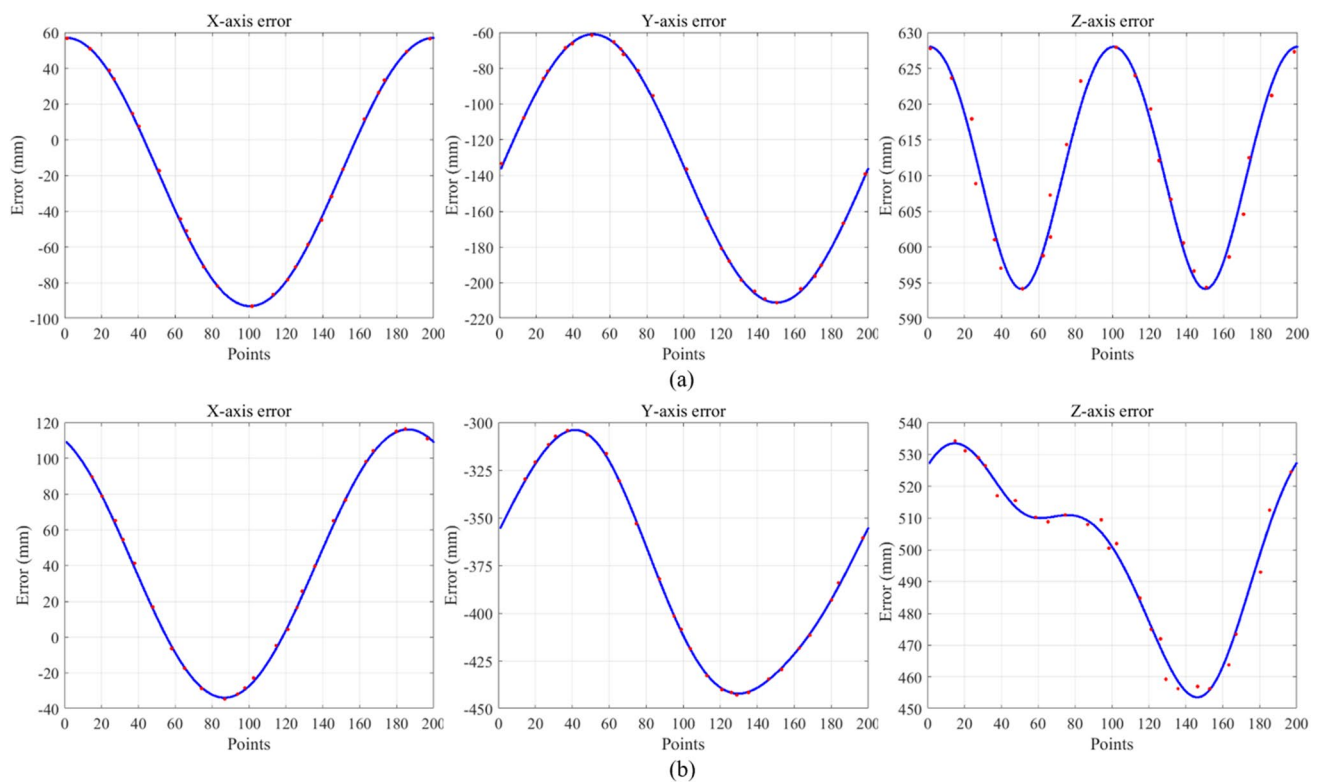


Fig. 23 X, Y, and Z-axis errors

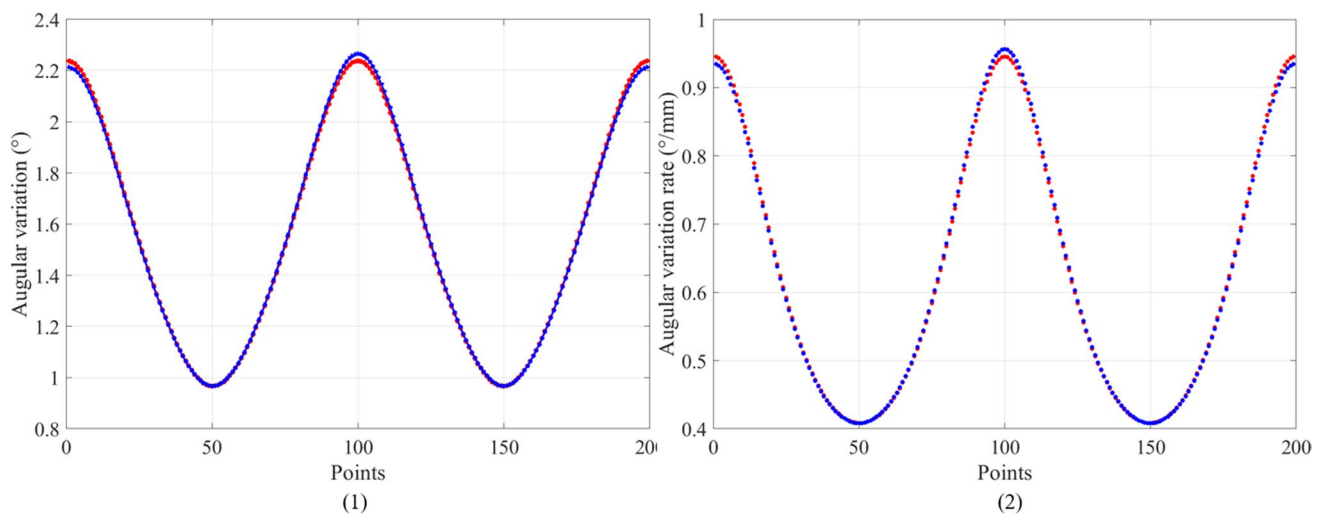


Fig. 24 Angular variation and angular variation rate

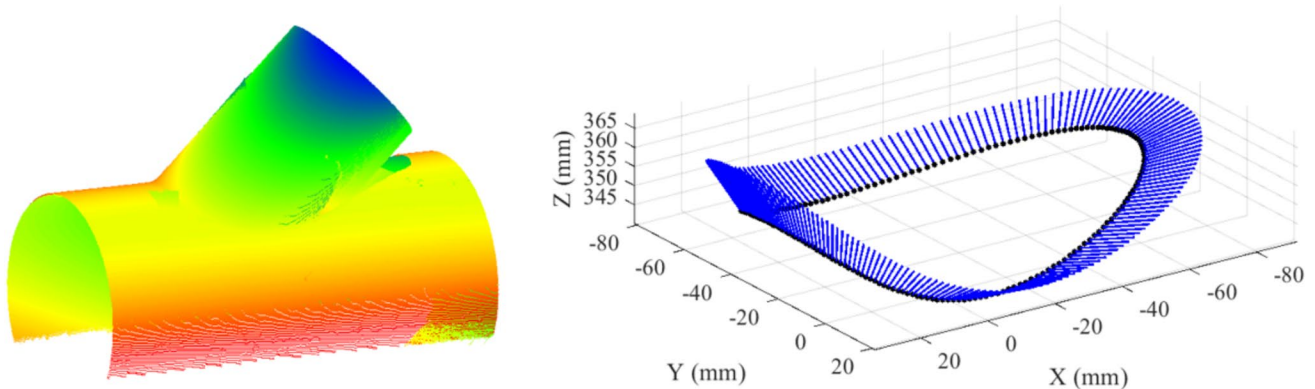


Fig. 25 Reconstruction result and path planning result

Author contribution All authors contributed to the study conception and design. Tianqi Wang: writing—review and editing, investigation, and data curation. Yifan Liang: validation, methodology, and writing—original draft. Xinqi Liu: writing—review and editing—and investigation. Zhigang Wang: data curation. All authors read and approved the final manuscript.

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Data availability The datasets generated and/or analyzed during the current study are not publicly available due to the proprietary nature of the data but are available from the corresponding author upon reasonable request.

Declarations

Competing interests The authors declare no competing interests.

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